

DSAA3073: Theories in Data Science

Week 7 Validation Sheet: Boosting II: Margins & Loss Minimization

Total Points: 130

Instructions:

- Write your answers clearly and legibly.
- Show all working for full credit.
- State any assumptions you make.
- For proofs, justify each step rigorously.
- This is a practice validation sheet. AI tools and references are allowed.
- Use this to assess your understanding before the actual test.

Part A: Multiple Choice (40 points)

Question 1. [5 marks] The margin of a training example (x, y) is defined as $\frac{y \cdot F(x)}{\|F\|_1}$. What does $\|F\|_1$ represent?

- a) The number of weak hypotheses used
- b) The sum of absolute values of hypothesis weights
- c) The maximum absolute value of hypothesis weights
- d) The training error of the combined classifier

Question 2. [5 marks] Which statement about the margin bound is TRUE?

- a) The bound always improves as the number of boosting rounds T increases
- b) The bound depends on the VC dimension of the combined classifier
- c) The bound depends on the margin distribution ρ_θ and the VC dimension of the weak learner
- d) Larger training error always leads to worse margin bounds

Question 3. [5 marks] For a correctly classified example with $y = +1$, $F(x) = 0.5$, and $\|F\|_1 = 5$, the margin is 0.1. Which statement is most accurate?

- a) This example has strong generalization potential
- b) This example is barely correct and may not generalize well
- c) This example is incorrectly classified
- d) The margin cannot be computed without knowing T

Question 4. [5 marks] Why is the exponential loss convex?

- a) It is a sum of convex functions
- b) The exponential function preserves convexity under composition with linear functions
- c) It upper bounds the 0-1 loss
- d) It has a unique global minimum

Question 5. [5 marks] In the coordinate descent interpretation, each round of AdaBoost:

- a) Updates all coordinates simultaneously
- b) Selects a random coordinate and optimizes it
- c) Greedily selects the coordinate with steepest descent and optimizes it
- d) Cycles through coordinates in a fixed order

Question 6. [5 marks] Compared to logistic loss, exponential loss is:

- a) More robust to outliers
- b) Less sensitive to misclassified examples
- c) More aggressive in penalizing errors
- d) Provides better calibrated probabilities

Question 7. [5 marks] What is ρ_0 in the margin distribution?

- a) The fraction of examples with zero margin
- b) The fraction of examples with negative margin (misclassified)
- c) The fraction of examples with positive margin
- d) The training error

Question 8. [5 marks] The formula $\alpha_t = \frac{1}{2} \ln \frac{1-\varepsilon_t}{\varepsilon_t}$ comes from:

- a) Maximizing the margin of the combined classifier
- b) Minimizing the exponential loss via line search
- c) Minimizing the training error directly
- d) Random initialization with geometric decay

Part B: Mixed Format Questions (90 points)

Question 9. [15 marks] Consider a training set with 6 examples. After T rounds of AdaBoost, we have $F(x) = \sum_{t=1}^T \alpha_t h_t(x)$ with $\|F\|_1 = 12$.

The predictions are:

Example	y_i	$F(x_i)$	Margin
1	+1	8	?
2	+1	3	?
3	-1	-6	?
4	-1	1	?
5	+1	10	?
6	-1	-2	?

- (a) [6 marks] Compute the margin for all examples. Which examples are misclassified?
- (b) [5 marks] Compute $\rho_{0.5}$. What does this tell us about the classifier's generalization?
- (c) [4 marks] If we run more boosting rounds, which examples would receive the highest weights? Why?

Question 10. [13 marks] (a) [7 marks] Prove that for a binary classification example, if the margin is positive, then the example is correctly classified. Conversely, if the margin is negative, the example is misclassified.

(b) [6 marks] Show that if all m training examples have margin $> \theta > 0$, then the training error is zero. Does the converse hold? Justify your answer.

Question 11. [15 marks] Consider the exponential loss $L(F) = \frac{1}{m} \sum_{i=1}^m \exp(-y_i F(x_i))$.

(a) [8 marks] Suppose we have a single training example (x, y) with $y = +1$, and we want to add a weak hypothesis h with weight α . Derive the value of α that minimizes $L(F + \alpha h)$ where $F(x)$ is the current prediction. Assume $h(x) = +1$.

(b) [7 marks] Generalize your result from part (a) to explain why AdaBoost's formula $\alpha_t = \frac{1}{2} \ln \frac{1-\varepsilon_t}{\varepsilon_t}$ is optimal for minimizing exponential loss over the entire training set.

Question 12. [12 marks] The margin bound includes a trade-off parameter θ :

$$P[H(x) \neq y] \leq \rho_\theta + O\left(\sqrt{\frac{d \log^2 \frac{m}{\theta^2} + \log\left(\frac{1}{\delta}\right)}{m}}\right) \quad (19)$$

(a) [6 marks] Explain intuitively why there is a trade-off as we vary θ . What happens to each term as θ increases?

(b) [6 marks] Suppose we have two classifiers on the same dataset:

- Classifier A: All examples have margin ≥ 0.6
- Classifier B: 95% of examples have margin ≥ 0.8 , but 5% have margin ≥ 0.1

Which classifier would you expect to have better generalization? Justify your answer using the margin bound perspective.

Question 13. [10 marks] (a) [5 marks] State three properties that make exponential loss suitable for AdaBoost. For each property, briefly explain why it matters.

(b) [5 marks] Despite these advantages, exponential loss has a significant weakness. What is it, and how do variants like LogitBoost address it?

Question 14. [15 marks] Consider three perspectives on why AdaBoost works:

1. **Algorithmic:** Adaptive reweighting focuses on hard examples
 2. **Optimization:** Coordinate descent on exponential loss
 3. **Margin:** Implicitly maximizes classification margins
- (a) [5 marks] Choose one specific aspect of AdaBoost (e.g., the weight update formula, the choice of α_t , or the selection of h_t). Explain this aspect from all three perspectives.
- (b) [10 marks] A skeptic claims: “The margin theory is just a post-hoc justification. AdaBoost was designed as an algorithmic procedure, and the margin maximization is an accidental byproduct.” Defend or refute this claim using evidence from the optimization perspective.

